

An adjoint-faithful Fourier neural operator for a double-gyre ocean: progress summary

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Abstract

This note summarizes progress, as of 25 August 2026, on training a Fourier neural operator (FNO) emulator of a MITGCM double-gyre configuration whose tangent-linear and adjoint sensitivities match the ocean model’s, not only its forecast trajectories. A 27.3-million-parameter, one-input/one-output FNO was trained from scratch on a weakly turbulent regime (S0) and then fine-tuned for a short second stage at a longer autoregressive horizon; the fine-tune improves essentially every forward forecast metric without trading away short-horizon skill. A preliminary adjoint comparison against a MITGCM/TAF ground-truth sensitivity map, however, finds that the trained emulator’s reverse-mode gradient is a poor match to the true adjoint: near-zero pattern correlation and amplitude errors of six- to thirty-eight-fold at leads of 10–90 days, concentrated in Jacobian error rather than trajectory drift. This finding motivates the adjoint-faithful training study now underway, which adds a forward perturbation-response term to the training objective without ever exposing the model to an adjoint value during training or selection. The response-data generation infrastructure and its integrity gates are built and passing; model training has not yet started.

1. The production forward emulator

1.1. Architecture and state

The overall map, state convention, and Fourier-operator backbone follow the FNO ocean-emulation architecture of Bire et al. [1], who developed and validated this class of emulator for the double gyre at $0.25^\circ \times 0.25^\circ$ resolution; the model described here adapts it to a one-input Markov map at one-degree resolution with the per-mode spectral normalization described below. The emulator is a single Markov operator $F_\theta : [x_t, S] \rightarrow x_{t+10}$ predicting the full 46-channel prognostic state ten days ahead on the MITGCM `tutorial_baroclinic_gyre` configuration (62×62, one degree): fifteen vertical

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levels each of zonal velocity U , meridional velocity V and potential temperature Θ , plus the free surface η . Five physical static channels (wind stress, wet mask, Coriolis parameter, grid spacing, SST relaxation target) and a two-channel sine/cosine position encoding are appended, for 53 lifted input channels. The state is deliberately kept closed — all fifteen levels of U , V , Θ , plus η — so that the map is a self-contained prognostic operator whose Jacobian can serve as a tangent/adjoint operator, which a reduced state would not support.

The network is a three-block Fourier neural operator, 32×32 retained Fourier modes, hidden width 128, with a parallel bias-free 3×3 local correction and six pointwise channel LayerNorms, 27,297,960 parameters in total (Table 1). Inside each spectral convolution the per-mode channel-mixing matrix is capped to unit spectral norm by two power iterations per forward pass [2], one-sided so that a mode already below the cap is left untouched and carries no gradient. The cap bounds one constituent of the block, not the full block Jacobian, so legitimate transient amplification through the skip connections, channel MLP and local branch remains representable.

Item	Value
Grid / interval	62×62 , $\Delta t = 10$ days, direct state (not residual)
Fourier modes	32×32
Hidden width	128
FNO blocks	3
Local branch	bias-free 3×3 , zero-initialized
Spectral normalization	per-mode $\sigma_{\max} \leq 1$, 1,632 modes, materialized at write
Parameters	27,297,960 (97.95% in the spectral convolutions)

Table 1: Core architecture of the production FNO, identical between the parent and the fine-tuned child.

1.2. Training, validation and inference

The *parent*, `model_c_production_1in_1out_spectralnorm_v1`, is trained from random initialization with an eight-term, fully dimensionless objective (state, increment, autoregressive rollout, spectral, boundary, pressure-gradient, continuity, barotropic-transport terms) on a six-call, 60-day autonomous rollout, 7,680 Adam steps — each step one optimizer update computed from an effective batch of eight 60-day training sequences — 3.21 h on one V100. The *child*, `.._ft90_v1`, is a short second stage — initialized from the parent, otherwise identical architecture, normalizer, and objective — extended to a nine-call, 90-day rollout at a constant, five-times-lower learning rate for 1,920 steps (37.5% of the parent’s exposure, 1.67 h). Every checkpoint is autoregressed 360 days from 102 pooled validation starts (34 evenly spaced starts in each of the three wind regimes S0/S1/S2) and selected by a rule with three requirements: short-horizon skill within 5% of the best candidate in every field; a bounded twin-perturbation growth rate (defined below); and, among the checkpoints meeting both, the smallest degradation of 90–360-day error relative to a train-only climatology baseline. Held inference uses 15 fixed S0 starts, never used for training or selection, autoregressed to day 2,000 against lead-matched MITGCM truth.

The growth rate is measured entirely within the trained FNO: its own initial state is perturbed by a small (1%) relative amount, and both the perturbed and unperturbed states are autoregressed through the *trained emulator itself*, with no MITGCM run at all, for 200 ten-day calls. Fitting the exponential rate at which the two trajectories separate over the later calls gives one number per checkpoint, capped at ≤ 1 per call, so a checkpoint whose own repeated map would amplify small differences into a runaway drift over a 2,000-day rollout is rejected before it ever reaches held inference. This is a different question from the adjoint comparison in Section 3. The growth rate asks only whether the trained FNO’s own nonlinear dynamics are self-consistently bounded when compared against themselves, iterated many times; it never involves MITGCM and says nothing about whether any single derivative of the emulator is physically correct. The adjoint study instead asks whether the *exact, infinitesimal* derivative of that same trained FNO, computed once by reverse-mode differentiation at a given lead, structurally matches MITGCM’s true derivative at the same point. A checkpoint can pass the first test and fail the second, and Section 3 shows that this is exactly what happens here: bounded, well-behaved own-dynamics do not imply a physically correct Jacobian.

1.3. Key results

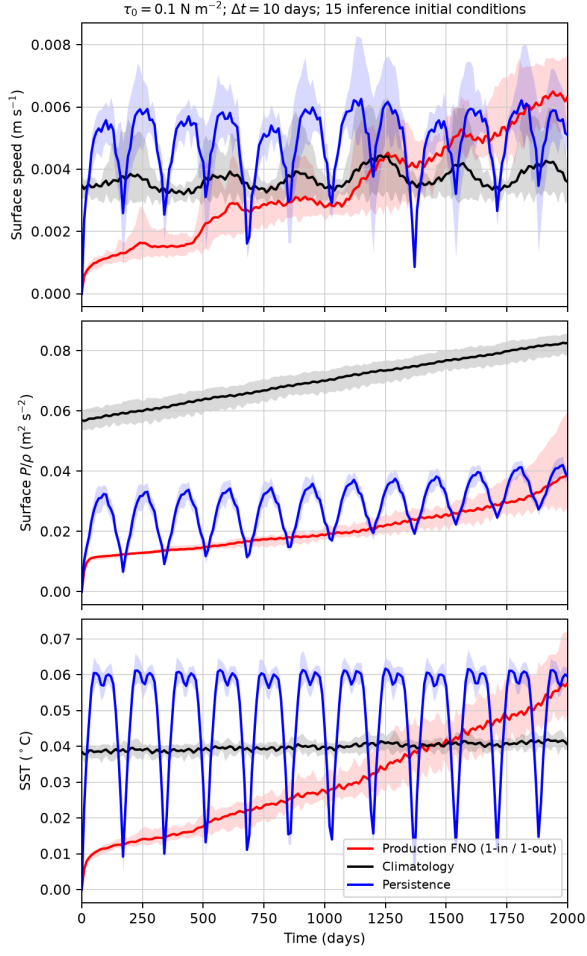
The fine-tune improves nearly every measured quantity relative to the parent, without the usual short-horizon trade-off (Table 2). Figures 1 and 2 show the pooled 15-member held inference through day 2,000 and the day-60/day-2,000 streamfunction anomaly for both models.

Metric	Parent	Fine-tuned (ft90)
10–90 d RMSE-AUC, surface pressure	0.579	0.267
Day-2,000 RMSE / climatology, SST	1.425 (worse)	0.924 (better)
Day-2,000 RMSE / climatology, pressure	0.478	0.248
Western boundary / interior anomaly ratio (truth 23.1)	5.06	8.16
Max. normalized amplitude through day 2,000 (target ≤ 8)	10.33	8.43
Twin-perturbation growth rate per call (target ≤ 1)	1.0132	1.0083

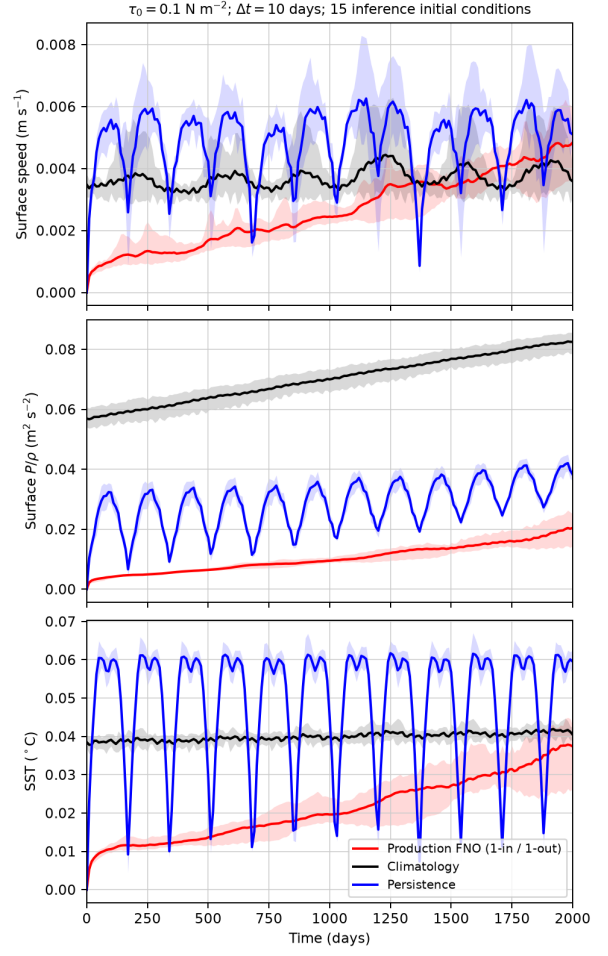
Table 2: Headline forward results, parent versus fine-tuned child, 15-member S0 held inference. Both still fail the declared day-2,000 amplitude and growth-rate gates; the fine-tune moves substantially toward them without reaching them.

Table 2 reads consistently in one direction: the fine-tune improves every entry. RMSE-AUC is the area under the RMSE-versus-lead-time curve over the stated window (10–90 days here) rather than the error at a single lead, so a lower value means uniformly better skill across the whole horizon, not just a better score on one particular day. The western boundary/interior anomaly ratio compares the day-2,000 streamfunction anomaly in the four-cell-wide western boundary band to the same quantity averaged over the interior; MITGCM’s value of 23.1 shows its transient anomaly is far more concentrated in the western boundary current than either emulator reproduces. Both models remain well short of that

concentration, though the fine-tune raises the ratio from 5.06 to 8.16 — about 1.6 times the parent’s value. The last two rows are the acceptance ceilings adopted for this project, and both are worst-case, not average-case, statistics. The normalized amplitude is measured in the same pointwise-normalized (z-score) coordinates the network is trained in: at every ten-day lead of every one of the 15 held-inference members, take the single largest absolute value across all 46 channels and all wet grid cells, then keep the largest such value over the whole 2,000-day rollout. A value of 8 means that, somewhere in space, in some channel, at some point in some member’s trajectory, the state reached eight training-climatology standard deviations from its pointwise mean; the target ≤ 8 caps that single worst excursion before the rollout is judged to have drifted into an unphysical regime, not the typical error level. The growth rate is the same twin-perturbation diagnostic described in Section 1, here reported as the worst value obtained from several independent validation starts rather than their average; the target ≤ 1 requires even the least favorable starting state to show no net amplification. Both models still exceed both ceilings, which is why day-2,000 behavior remains an open problem even though every other entry in the table improves.

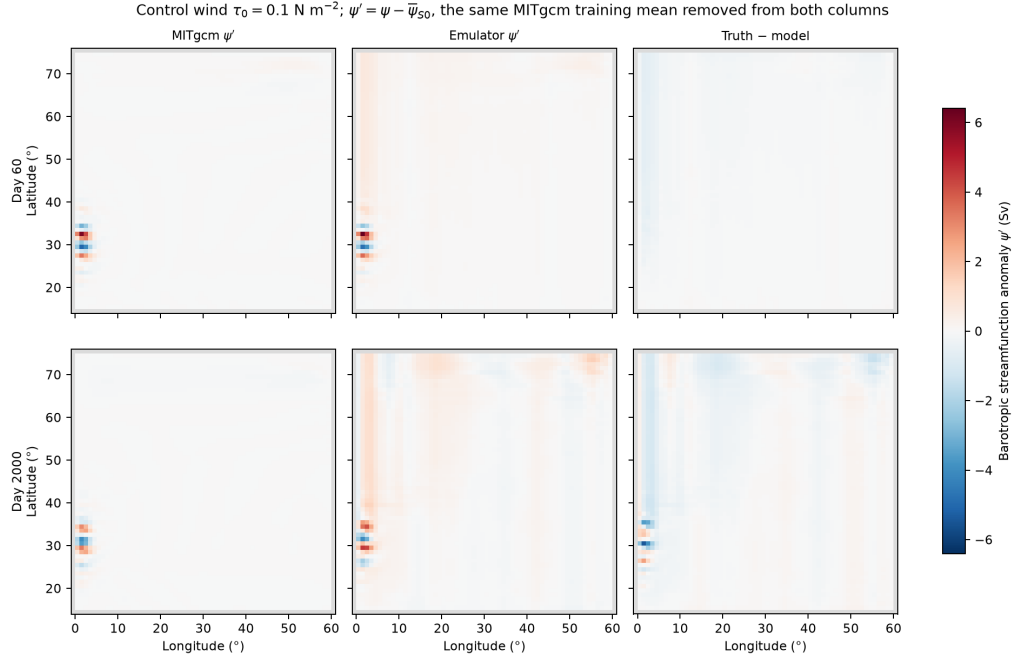


(a) Parent

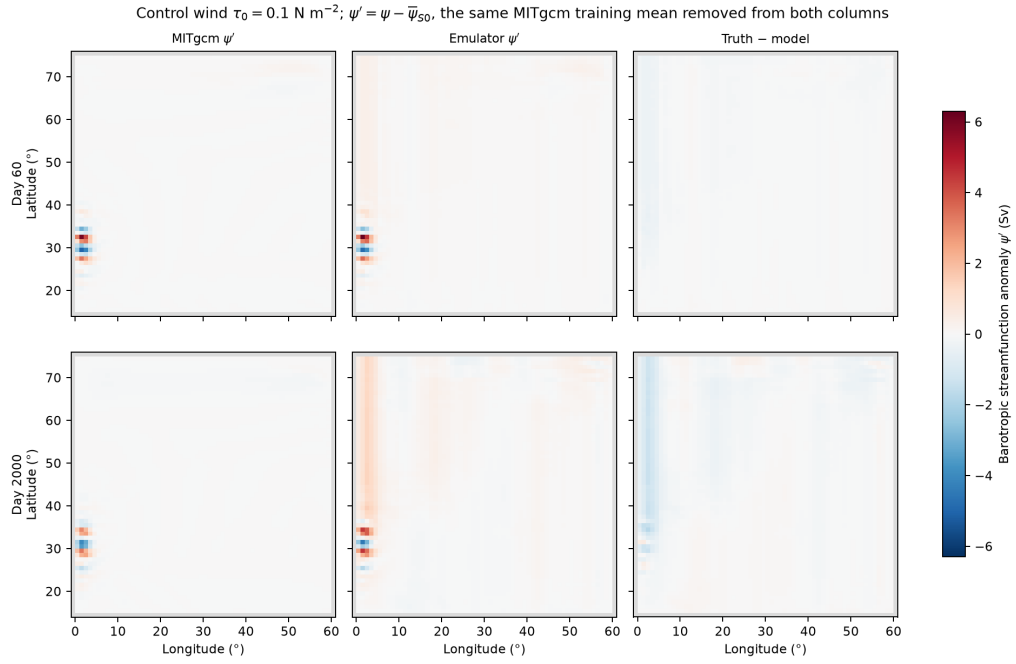


(b) Fine-tuned (ft90)

Figure 1: Pooled 15-member held-inference RMSE against MITGCM truth, $\Delta t = 10 \text{ d}$, days 0–2,000, S0. The fine-tune’s error grows more slowly in all three fields but has not plateaued by day 2,000.



(a) Parent



(b) Fine-tuned (ft90)

Figure 2: Barotropic streamfunction anomaly about the training-mean circulation, day 60 and day 2,000, S0. Both models keep the gyre identifiable with a sharp western boundary current at day 2,000; transient amplitude remains $\sim 1.7\text{--}1.8\times$ too large relative to truth in both.

2. Why a weakly turbulent double gyre

All results above use the S0 regime, the control wind amplitude ($\tau_0 = 0.100 \text{ N m}^{-2}$) matching Bire et al.'s [1] original configuration; S1 and S2 are a weaker and a stronger wind perturbation (0.075 and 0.125 N m^{-2}). A weakly turbulent double gyre minimizes chaotic and multiscale complexity, which gives a clean benchmark to isolate whether the FNO has learned the correct tangent and adjoint dynamics before moving to a harder, genuinely eddying regime. Figure 3 supports this directly: two MITGCM pickups differing by a 10^{-3} relative perturbation never separate from each other over 25 simulated years, sitting three to four decades *below* the control trajectory's own decadal drift throughout. S0 does not exhibit butterfly-effect sensitivity to initial conditions at this amplitude, so a discrepancy between the emulator and truth over a long rollout is model error, not chaotic divergence being amplified — exactly the property needed to make a Jacobian/adjoint comparison interpretable before repeating it on the genuinely turbulent 0.25° S0/S1/S2 campaign.

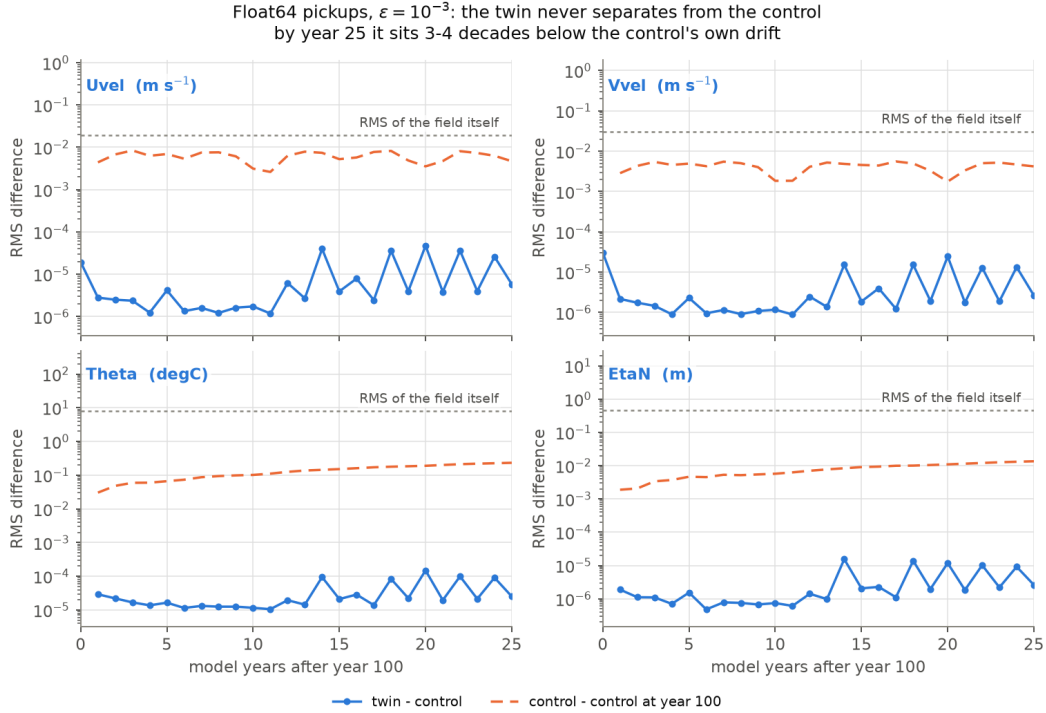


Figure 3: Twin-pickup divergence in S0: RMS difference between a control trajectory and a $\varepsilon = 10^{-3}$ perturbed twin, both float64 MITGCM, over 25 model years. The twin-control difference (solid) never approaches the control's own year-to-year drift (dashed) or the field's own RMS (dotted), confirming S0 is not chaotic at this perturbation scale.

3. Preliminary adjoint study

3.1. Objective and setup

The comparison targets a scalar sea-surface-height objective and its gradient with respect to the initial surface,

$$S_T[j, i] = \frac{\partial J_T}{\partial \eta(j, i, t_0)}, \quad T \in \{10, 20, 30, 90\} \text{ days},$$

evaluated two ways on the identical S0 trajectory and window: through MITGCM differentiated by TAF 6.8.11 (reference, ~ 20 min per lead), and through reverse-mode automatic differentiation of the trained, one-input production FNO (test, seconds). The primary objective is a point functional at a fixed western-boundary cell p^* , referenced to the domain-mean:

$$J_{\text{point}} = \eta(p^*, T) - \langle \eta(T) \rangle_A,$$

with a secondary five-point meridional Gaussian-smoothed variant, J_{kernel} , used as a robustness check. Before any comparison number below is trusted, both producers of S_T are independently verified. The MITGCM/TAF gradient is checked against direct finite differences of the forward model over a spread of perturbation sizes, confirming an interior error minimum rather than a monotonic trend against either round-off or truncation error; it is exactly reproducible run to run, exactly zero at every land cell, and reproduces the analytically known conservation of the basin-mean sea level. The FNO’s reverse-mode gradient is checked against forward-mode automatic differentiation of the same network, against direct finite differences of the trained operator itself, and confirmed to satisfy the adjoint identity $\langle v, Ju \rangle = \langle J^T v, u \rangle$ to near machine precision. Both sides pass every one of these checks at every lead before any MITGCM-versus-FNO comparison below is reported.

3.2. Headline result

The comparison, run against the fine-tuned child (the frozen parent has not yet been passed through the current pipeline), finds that forecast skill and adjoint fidelity are not the same thing. Pattern correlation between the emulator’s and MITGCM’s sensitivity maps is essentially zero at every lead (0.06, -0.01 , 0.01, 0.02 at 10/20/30/90 days), and the emulator’s amplitude is wrong by a factor of 38 at 10 days, decaying to 6.6 at 90 days (Figure 5). The spatial mismatch is qualitative, not a fine-scale correction on an otherwise-correct field: at 90 days the true sensitivity is smooth and concentrated near the western boundary, while the emulator’s is fine-grained and sign-incoherent over most of the basin (Figure 4). The sensitivity values themselves are small in an absolute sense — $O(10^{-5})$ in Figure 4 — because S_T is a dimensionless influence coefficient: the fraction of a unit perturbation at one grid cell that survives at the single objective cell T days later. That influence is spread over the domain’s $\sim 3,600$ wet cells, so no individual cell is expected to carry an $O(1)$ share; values several orders below one are the physically expected scale here, not evidence of a weak or unreliable signal. The emulator also destroys the analytically conserved basin-mean SSH mode, whose exact answer is amplitude ratio 1.0 at every lead: the emulator recovers 41% of it at 10 days and 0.6% at 90 days.

Two variants of the emulator’s sensitivity separate where the error comes from. S_{forced} linearizes the emulator about the true MITGCM trajectory at every ten-day step, so it isolates whether the emulator’s own ten-day Jacobian is right, independent of where its autoregressive rollout would actually have gone; S_{free} instead linearizes about the emulator’s own autoregressive trajectory, which is what an operational user of the deployed model actually gets. The two agree closely — pattern correlation 0.996 at 90 days (Figure 5) — so the mismatch against MITGCM is overwhelmingly Jacobian error rather than trajectory drift: the emulator’s forecast trajectory is close to correct, but its local derivative along that trajectory is not. That is encouraging for the training intervention in Section 4, since it means the quantity that intervention directly supervises is the one that is wrong.

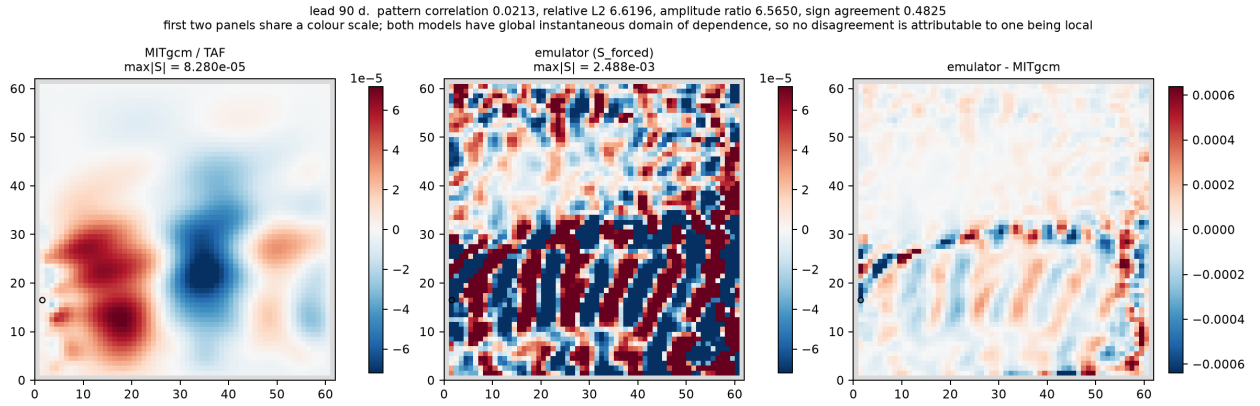


Figure 4: Sensitivity map S_{90} at 90 days: MITGCM/TAF (left), the fine-tuned FNO (centre), and their difference (right), same color scale on the first two panels. Pattern correlation 0.021, amplitude ratio 6.57.

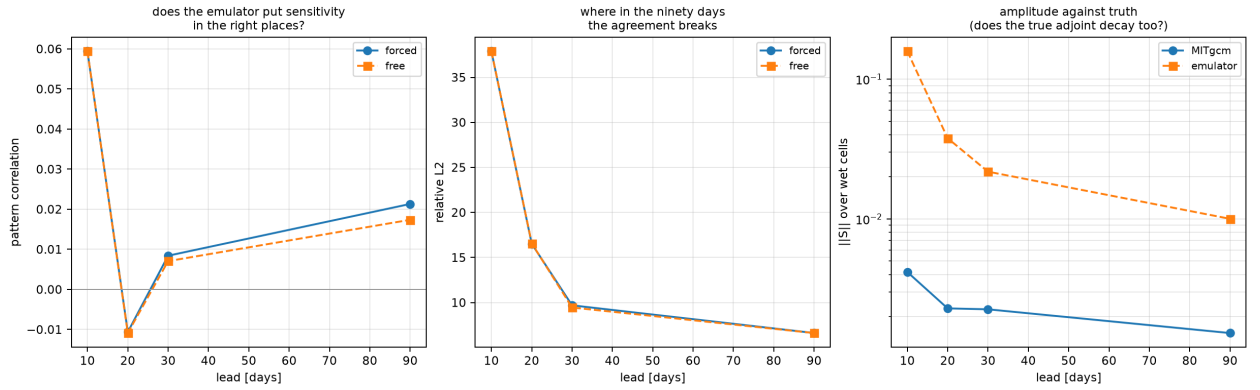


Figure 5: Lead sweep of the point-objective comparison. Left: pattern correlation of S_{forced} and S_{free} (defined in the text) against MITGCM, both near zero beyond lead 10. Centre: relative L_2 error, decaying with lead but never small. Right: sensitivity amplitude against truth — the MITGCM adjoint decays gently while the emulator’s over-damps by an order of magnitude by day 90.

4. Overview of the adjoint-faithful response training plan

The forward-skill/adjoint-fidelity gap in Section 3 motivates a targeted intervention: supervise the FNO on finite-difference directional derivatives of the true MITGCM flow map, without ever letting an adjoint value influence training or model selection. For a direction v and amplitude ϵ , the core identity relates forward finite differences to the Jacobian-vector product,

$$\frac{M_k(x + \epsilon v) - M_k(x - \epsilon v)}{2\epsilon} = DM_k(x) v + O(\epsilon^2),$$

with the analogous expression for the emulator F_θ . Matching these forward responses constrains Jacobian-vector products for the sampled directions; by duality, for any cotangent w , $\langle (DF_k^T - DM_k^T)w, v \rangle = \langle w, (DF_k - DM_k)v \rangle$, so a better-matched forward response implies a better-matched adjoint *projection* onto the sampled direction span, though not the full Jacobian in unsampled directions — which is exactly what the blind adjoint test at the end of the study is designed to check.

The response loss. The intervention is built directly on the identity above. A large forward-only MITGCM dataset is generated by integrating paired plus/minus perturbed states: at each sampled anchor state x_q , input group $h(q) \in \{U, V, \Theta, \text{SSH}\}$, direction v_q , and signed amplitude $s\epsilon_q$ with $s \in \{-1, +1\}$, MITGCM is integrated from $x_q + s\epsilon_q v_q$ and from the unperturbed x_q , and the normalized, signed response

$$r_{M,q,k}^s = \frac{\mathcal{N}[M_k(x_q + s\epsilon_q v_q)] - \mathcal{N}[M_k(x_q)]}{s\epsilon_q}$$

is recorded at each ten-day lead k out to a role-dependent horizon, in the same pointwise-normalized coordinates $\mathcal{N}(\cdot)$ the emulator is trained in. The analogous quantity $r_{F,q,k}^s$ is computed from the emulator on the same edited initial states. A new model, F'_θ , is trained from random initialization with the same architecture, data split, optimizer schedule, and eight-term forward objective L_{nominal} as the production emulator of Section 1, plus exactly one additional term added on a predeclared subset of optimizer steps,

$$L_{\text{total}} = L_{\text{nominal}} + \lambda_{\text{resp}} L_{\text{response}}.$$

Writing $g \in \{U, V, \Theta, \text{SSH}\}$ for an output group and $d_{h,g,k}^2$ for the response-training-data mean-square response of output group g to a perturbation of input group h at lead k ,

$$L_{\text{response}} = \text{mean}_q \frac{1}{8} \sum_{s \in \{-1, +1\}} \sum_g \frac{\text{mean}_{c \in g, \text{wet}} (r_{F,q,k}^s - r_{M,q,k}^s)^2}{d_{h(q),g,k}^2},$$

averaged over the sampled directions q and, for the subset of directions extended beyond ten days, over lead k as well. Dividing by $d_{h,g,k}^2$ puts every input/output pair on a comparable, unitless scale regardless of its physical units or channel count, so no single group can dominate the loss simply by having larger natural units or more vertical levels. No ordinary

state loss is placed on the perturbed trajectories themselves — the signed response is the only quantity supervised — and no adjoint output of any kind, from MITGCM/TAF or from the emulator, enters L_{response} or any other part of training: the response term is the only difference between this model and the existing production training recipe.

Data scale. Generating this dataset is itself a substantial MITGCM campaign. Perturbing U , V , Θ , and sea-surface height at many spatial locations and times across the three wind regimes, both signs, mostly to a ten-day lead with a smaller subset carried to 60 or 90 days, totals 57,750 model-days (160.4 model-years) of forward-only integration across the training, validation, and a held-out blind set. As above, no MITGCM adjoint or TAF output is used anywhere in generating this data or in training: every quantity feeding L_{response} comes from ordinary forward integrations.

Direction and status. The plan is staged so that every design choice — how large a perturbation to use, how to weight the response term, which checkpoint to keep — is frozen using forward evidence alone, before the trained model is ever compared against MITGCM’s adjoint. Concretely: first calibrate, with a small forward-only pilot, how large a perturbation can be while the response still behaves linearly; then generate the full training/validation/blind response dataset at that calibrated amplitude; then train the new model, choosing its response weight and checkpoint using only forward-forecast skill and held-out response error, while requiring it to match the existing production emulator’s ordinary forecast performance; and only once all of that is locked in, evaluate the trained model’s adjoint against MITGCM/TAF as in Section 3, to see whether forward-response supervision closes the gap found there. This ordering is deliberate: it means a good- or bad-looking adjoint result can never feed back into how the model was built or chosen.

Next steps

The immediate next step is a small forward-only pilot that perturbs each of U , V , Θ , and sea-surface height at a handful of locations and candidate amplitudes, to fix one amplitude per variable before large-scale data generation begins. From there the plan moves to generating the full response dataset, training the response-aware model and choosing its response weight, and finally revisiting the Section 3 comparison against MITGCM/TAF with the new model, to test whether this forward-only supervision is enough to fix what Section 3 found.

References

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- [2] M. McCabe, P. Harrington, S. Subramanian, J. Brown, Towards stability of autoregressive neural operators, *arXiv preprint arXiv:2306.10619* (2023).